

Michael Nyale Munga

Backend Software Engineer | Operational Data Analytics | Supply Chain & ERP Systems

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SUMMARY

Backend engineer who started in procurement and inventory control and now builds the systems and analysis behind business operations. Most recently worked on Java/Spring Boot microservices for live SACCO and Agency Banking platforms at E&M Technology, while engineering analytics projects covering forecasting, risk scoring, root-cause analysis, and dashboards. Skilled in Python, React/Next.js, and Odoo. Having operated directly on the procurement and inventory side, I understand what the data structures and approval workflows behind these enterprise systems actually represent in production.

SKILLS & COMPETENCIES

AREA	TECHNICAL & FUNCTIONAL STACK
Backend	Java, Spring Boot, Spring Cloud (Eureka), Resilience4j, Python, Flask, FastAPI, REST API design, JWT/RBAC
Frontend	Next.js (App Router), React, Angular, TypeScript, Tailwind CSS, Recharts
Data & Analytics	Pandas, SQL, ETL & validation (Great Expectations), demand and stockout forecasting, statistical testing, root-cause analysis, Streamlit
ERP & Supply Chain	Odoo (ORM, XML views, REST endpoints, module architecture), inventory control, reorder-level planning, supplier performance analysis, public-sector procurement compliance
DevOps & Testing	PostgreSQL, Docker, Docker Compose, Celery, Redis, Git/GitHub, WireMock, Testcontainers, M-Pesa Daraja API

PROFESSIONAL EXPERIENCE

Backend Engineer Jun 2026 – 18 Sep 2026
E&M Technology | Nairobi, Kenya

- Engineered Java/Spring Boot microservices for SACCO lending and Agency Banking platforms within an agile team operating through structured pull requests and strict code reviews.
- Contributed to the loan-management services of a SACCO lending system and to the transaction-processing and account-management services of an Agency Banking platform on a shared Spring Boot, Spring Cloud, and Eureka architecture.
- Participated in a multi-service refactor that placed Core Banking vendor calls behind a single connector, isolating third-party API changes, and implemented circuit breakers and idempotent postings using **Resilience4j** and **Anti-Corruption Layer (ACL)** patterns to prevent duplicate postings during network retries.
- Built automated test coverage for the identity service, encompassing controller tests, JWT security validations, method security checks, and integration tests running against containerized dependencies using **WireMock** and **Testcontainers**.
- Diagnosed and resolved two production-critical issues: rectified a misconfiguration leaving the shared global exception handler disabled in production, and resolved multi-service startup failures stemming from database migration and library dependency collisions.
- Delivered responsive frontend features in **Angular/TypeScript** alongside core backend service implementations.

Stores Assistant: Procurement & Inventory Control Aug 2024 – Jun 2025
Kibaoni Farm Care Agrovet | Kilifi, Kenya

- Achieved 98% inventory accuracy** through systematic physical and digital stock reconciliations across agricultural inputs.
- Managed end-to-end procurement cycles and maintained strategic supplier relationships to secure consistent product availability.
- Synthesized stock variance, dynamic reorder thresholds, and supplier-performance metrics into structured decision-support reports used by executive leadership for purchasing decisions.

Procurement Trainee Feb 2024 – May 2024
Kilifi Water and Sewerage Company | Kilifi, Kenya

- Administered statutory Local Purchase Orders (LPOs) and conducted market price surveys within a regulated public-sector framework.
- Maintained rigorous compliance records and audit trails, building foundational domain knowledge of approval hierarchies and procurement regulations.

PROJECTS

KEMSA Healthcare Supply Chain Intelligence Platform

Python, FastAPI, Next.js, SQL

Modelling & ML Lead in a team of five | Simulated dataset covering Kenya's 47 counties

- Built a stockout forecasting engine that projects **30/60/90-day stockout dates across 1,500 facility-commodity pairs**, using recursive depletion modelling rather than per-pair Prophet/ARIMA to maintain low query latency.
- Designed an explainable county debt-risk score (factoring days overdue, debt amount, and payment history) with Low/Medium/High tiers and plain-language reasoning.
- Constructed a Next.js frontend with server-side middleware enforcing role and county/facility scopes on every route.
- Shipped national, county, and facility-level dashboards complete with a transfer approve/reject workflow, debt trend charts, and CSV exports.

Depot Flow Ops: Throughput & Turnaround Analytics

Python, Great Expectations, React

- Constructed an automated ETL pipeline incorporating Great Expectations validation gates feeding a deployed operational dashboard across five regional fuel depots.
- Identified that **weighbridge-to-bay waiting time (92–106 min) substantially exceeded actual loading time (59 min)**, isolating the primary driver of demurrage expenses.

Pipeline Operations Analytics

Python, Pandas, SQL, Streamlit

- Engineered a multi-source data integration pipeline combining telemetry sensor logs, weather API feeds, and an operational SQLite database.
- Statistically evaluated a headline correlation ($r = -0.78$) using leave-one-out, Spearman rank, and Fisher z-transform tests, demonstrating that a single data outlier was driving the result.
- Executed root-cause analysis (Pareto charting, 5-Whys methodology) that traced the largest proportion of network downtime directly to pump seal failures at a specific pumping station.
- Delivered an interactive Streamlit safety dashboard for monitoring incident telemetry featuring real-time KPIs, multi-criteria filtering, alerting logic, and CSV data exports.

The Vintage Closet: Full-Stack E-Commerce Platform

Flask, React, PostgreSQL, Docker

- Independently developed and shipped a full-stack platform consisting of a React/Tailwind customer storefront, administrative console, and a Flask REST API managing auth, order workflows, and inventory levels.
- Integrated **M-Pesa Daraja STK Push** for automated mobile payments with asynchronous callback handling.
- Implemented JWT authentication, RBAC, API rate limiting, and asynchronous background task processing using Celery and Redis.

Odoo Custom Addon & Headless API

Python, Odoo 18, XML, PostgreSQL, React

- Developed a custom Odoo 18 addon from scratch (data models, relational XML views, menus, and automated business logic).
- Exposed modular REST endpoints secured with Bearer token authentication to interface directly with a decoupled React frontend, utilizing Odoo as a headless backend engine.

EDUCATION

Software Engineering Bootcamp

Feb – Jul 2025

Moringa School | Nairobi, Kenya

- Intensive full-stack software engineering program focused on system design, data structures, REST API architectures, and enterprise software engineering.

B.A. Business Management (Supplies & Procurement Management)

2020 – 2024

Mount Kenya University | Kenya

- Specialized coursework in Supply Chain Operations, Public Procurement Procedures, Contract Administration, Inventory Valuation, and Business Law.

CERTIFICATIONS

- **African Management Institute (AMI):** Performance Management, Problem Solving, Goal Setting, Emotional Intelligence

REFERENCES

- Available upon request.